

Credit Card Default Auto-ML Report — xgb (sub)

 2026-05-20 23:55:05 UTC

모형 **xgb**

roc_auc = **0.7777**

BEST(catboost) = 0.7796

Δ vs best = -0.0019

1 점수 요약 (본 모형 vs best)

metric	xgb (현재)		catboost (BEST)		Δ(Test)
	Train	Test	Train	Test	
roc_auc	0.8222	0.7777	0.8099	0.7796	-0.0019
ks	0.4841	0.4265	0.4660	0.4362	-0.0097
lift	3.3189	3.1123	3.4037	3.1274	-0.0151
accuracy	0.8278	0.8195	0.8308	0.8177	+0.0018
precision	0.7078	0.6723	0.7133	0.6589	+0.0134
recall	0.3773	0.3587	0.3931	0.3640	-0.0053
f1	0.4922	0.4678	0.5069	0.4689	-0.0011
pr_auc	0.6174	0.5548	0.6194	0.5588	-0.0040

2 Feature Importance — xgb

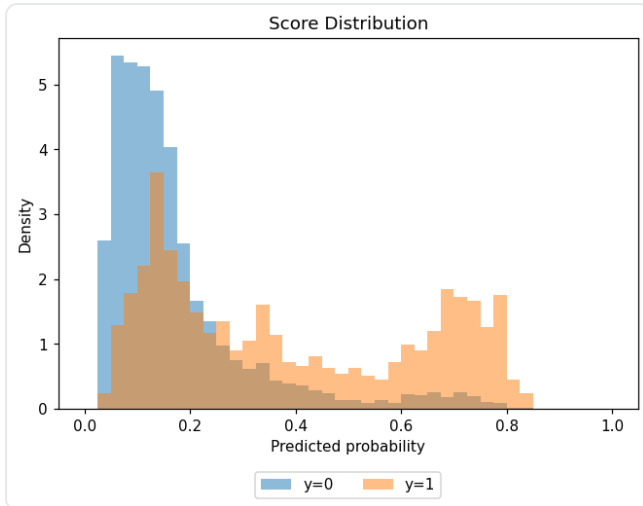
relative · top 19 · 전체 분포는 feature_importance.csv

Rank	Feature	Importance	Rank	Feature	Importance
1	PAY_0	0.4916	16	SEX	0.0114
2	PAY_2	0.1904	17	BILL_AMT5	0.0110
3	PAY_3	0.0443	18	BILL_AMT6	0.0107
4	PAY_4	0.0345	19	AGE	0.0101
5	PAY_5	0.0305			
6	PAY_6	0.0280			
7	PAY_AMT2	0.0199			
8	LIMIT_BAL	0.0188			
9	PAY_AMT3	0.0171			
10	BILL_AMT1	0.0167			
11	PAY_AMT1	0.0151			
12	BILL_AMT2	0.0132			
13	PAY_AMT4	0.0130			
14	PAY_AMT5	0.0122			
15	PAY_AMT6	0.0115			

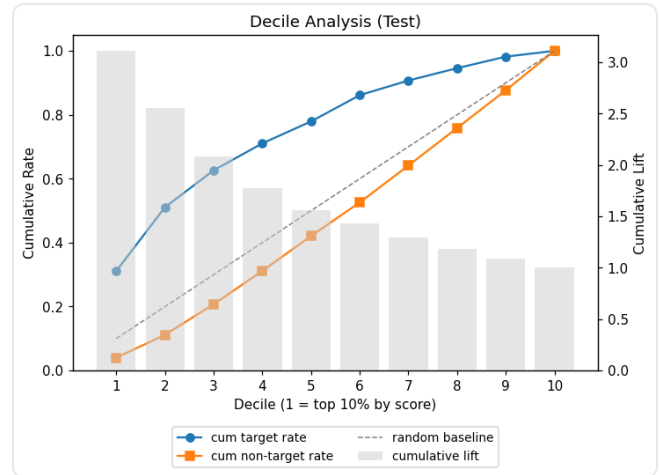
3 모델 상세 — xgb

점수 분포 · 10-분위 차트

Score Distribution by Label



Decile Analysis Chart



Decile	Score Range	Count	Target	Non-Target	Cum Target	Cum NonTarget	Cum Count	Cum %
1	0.585 ~ 0.850	600	413	187	413	187	600	10.0%
2	0.328 ~ 0.585	600	265	335	678	522	1200	20.0%
3	0.223 ~ 0.328	600	153	447	831	969	1800	30.0%
4	0.175 ~ 0.223	600	112	488	943	1457	2400	40.0%
5	0.147 ~ 0.175	600	91	509	1034	1966	3000	50.0%
6	0.127 ~ 0.147	600	109	491	1143	2457	3600	60.0%
7	0.105 ~ 0.127	600	61	539	1204	2996	4200	70.0%
8	0.083 ~ 0.105	600	51	549	1255	3545	4800	80.0%
9	0.060 ~ 0.083	600	48	552	1303	4097	5400	90.0%
10	0.032 ~ 0.060	600	24	576	1327	4673	6000	100.0%

Confusion Matrix threshold = 0.5

	Pred 0	Pred 1
True 0	4441	232
True 1	851	476

4 변수 선택 (Stability Selection)

base: lasso

부분표본 80회

임계값 0.60

채택 19 / 23

표 노출 23 / 전체 23 (전체 분포는 메인 리포트의 feature_selection.csv)

Feature	Selection Frequency		채택
LIMIT_BAL		100.0%	✓ 채택
AGE		100.0%	✓ 채택
PAY_0		100.0%	✓ 채택
PAY_2		100.0%	✓ 채택
PAY_3		100.0%	✓ 채택
PAY_4		100.0%	✓ 채택
BILL_AMT1		100.0%	✓ 채택
PAY_AMT1		100.0%	✓ 채택
PAY_AMT2		100.0%	✓ 채택
PAY_AMT3		100.0%	✓ 채택
PAY_AMT6		100.0%	✓ 채택
PAY_5		98.8%	✓ 채택
BILL_AMT5		98.8%	✓ 채택
PAY_6		97.5%	✓ 채택
PAY_AMT4		96.2%	✓ 채택
PAY_AMT5		95.0%	✓ 채택
SEX		91.2%	✓ 채택
BILL_AMT6		88.8%	✓ 채택
BILL_AMT2		71.2%	✓ 채택
BILL_AMT3		56.2%	제외
BILL_AMT4		28.7%	제외
EDUCATION		21.2%	제외
MARRIAGE		15.0%	제외

5 학습 설정 + 튜닝 — xgb

키	값
tuned	YES n_trials = 15 , best_value = 0.7855
best_iteration (avg over folds)	175
params (최종)	{'objective': 'binary:logistic', 'eval_metric': 'auc', 'tree_method': 'hist', 'n_estimators': 800, 'learning_rate': 0.026000059117302653, 'max_depth': 7, 'subsample': 0.6563696899899051, 'colsample_bytree': 0.9208787923016158, 'min_child_weight': 1.6709557931179373, 'reg_alpha': 7.620481786158549, 'reg_lambda': 0.08916674715636537, 'gamma': 0.993578407670862}